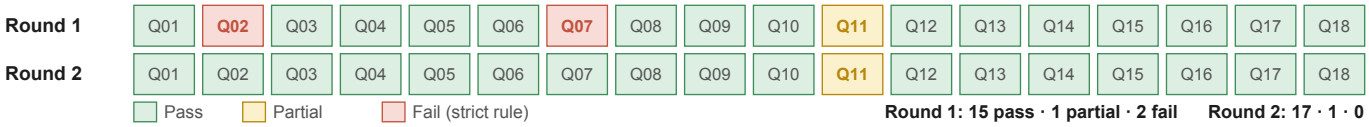
Evaluation results

18 test questions run blind and scored against an answer key computed from the raw tables, with one improvement round.

01 Summary



Round 1	15 pass · 1 partial · 2 fail	Every question in a fresh session. Q02 in Claude Code; the other 17 in Codex.
Round 2	17 pass · 1 partial · 0 fail	After skill updates: Q02 revised after review and rerun blind; Q07 and Q11 rerun; Q01 and Q09 rerun to check nothing broke; other results carry over.
Fixed	Q02, Q07	Overstated certainty and mixed periods (Q02); the start-up month used as a normal baseline (Q07).
Still open	Q11	The start-up rule didn't trigger for a trend description; recorded as the next fix.
Strongest	Ambiguity and traps	Asked or stated a clear choice when the question was open (Q03, Q10, Q15, Q18), declined the unanswerable question (Q16) and caught both planted data issues (Q12, Q13).

Question type	Questions	Result	What happened
Simple numbers and filters	Q01 Q05 Q08 Q17 Q18	**All pass**	Exact to the user and the cent
Vague questions	Q03 Q10 Q15 Q18	**All pass**	Asked, or stated a clear choice, before picking a metric, period or meaning
Unanswerable question	Q16	**Pass**	Declined and named the data that's missing
Data-quality traps	Q12 Q13 Q14	**All pass**	Caught both planted issues; didn't claim to have detected a reviewed one
Comparisons and lifecycle	Q04 Q06 Q09	**All pass**	Right time basis; groups reconcile to the total
Cohort conversion	Q07	**Fail → Pass**	Treated April, the start-up month, as a normal baseline; a new skill rule fixed it
Funnel drop-off	Q11	**Partial**	Right answer, but read April's start-up as a funnel decline; the rule didn't trigger
GM investigation	Q02	**Fail → Pass**	v1 right on numbers but too firm on some conclusions; after three certainty rules, v2 and a blind rerun both passed

02 How the evaluation worked

- **Questions:** 18 scenarios covering every type the brief lists: simple aggregates, filtered queries, time comparisons, cohorts, ambiguous and unanswerable questions, data-quality traps and a multi-step investigation (investigation-paths.md).
- **Independent references:** expected values were computed with the metric registry's SQL on the **raw source tables only**, never on the Silver and Gold tables the agent uses. Q02 used reviewer source checks; other scenarios used approved answer-key batches. In Q03, the evaluator chose weeks outside the initial key, so their values were computed after the run using the same method. The reference queries are kept as evidence logs.
- **Who scored:** Claude Code scored every run against the key. The GM report (Q02 v1) was also reviewed independently with ChatGPT, and the project owner decided the borderline calls.
- **Blind runs:** one fresh session per question with the same prompt; the agent was told not to read evaluation files, earlier reports or the data generator's notes. When it asked a question, the reply named a choice only (e.g. "Monthly, Q2 vs Q3 2026").
- **Scoring:** exact counts and cents; rates within 0.1 point. Behaviour counts too: a conclusion firmer than the evidence, a silent choice of period or definition, or a missing required caveat fails even when the numbers are right.
- **Improvement loop:** each failure became a general rule in the shared skill (never a question-specific answer); failed questions and two passing ones were rerun in fresh sessions on the updated version.
- **Snapshot:** pockettoons.sqlite (SHA-256 8d39e78b…790b), build 3a0dbed2 with 122 of 122 layer checks passed; April–September 2026, complete UTC months.

03 Every test: expected vs actual

R1 = round 1, R2 = round 2. Each folder link opens the run's scoring record, the agent's unedited output and its evidence logs on GitHub.

Test	Expected (answer key)	Agent result	R1	R2
Q01 · Simple aggregate; time comparison *How did monthly active users change across the available months?* Runtime: Codex Q01/codex-run-001 Q01/codex-run-002	MAU Apr–Sep: 7,937 · 9,134 · 9,976 · 10,138 · 10,871 · 11,447; +3,510 (+44.2%). Complete UTC months, distinct users; unidentified sessions excluded and not counted as people; metric version stated; months never summed.	Every month exact, with month-on-month changes. Stated mau v1.1, excluded unidentified sessions (2,286–3,703 a month) and recounted from Silver, matching Gold. Round 2 identical.	Pass	Pass
Q02 · Multi-step investigation (GM question) *MAU is up 20% QoQ but revenue is flat. What's happening and what should we do?* Runtime: Claude Code Q02/run-001 Q02/run-002-blind	Avg MAU 9,015.7 → 10,818.7 (+20.0%); net revenue $30,145.00 → $30,166.97 (+0.07%); paying users ~1,036 flat. Growth from new users (paid social +1,841); cohorts compared on complete windows; patterns stated as associations, not causes.	v1: numbers exact and story right, but some conclusions too firm (cause stated, two hypotheses rejected too firmly, data-quality impact called "none"), conversion periods mislabelled and a monthly/quarterly mix-up. v2 after review: all points addressed. Blind rerun on the updated skill: numbers exact, review points fixed; minor wording issues.	Fail	Pass
Q03 · Decomposition; vague period *Which acquisition sources contributed most to the change in active users?* Runtime: Codex Q03/codex-run-001	Ask for or state the period. For the weeks the evaluator chose when asked (27 Jul–2 Aug vs 17–23 Aug; values computed after the run): WAU 5,914 → 5,831; organic −213, paid social +182, paid search −52, unknown 67 → 67 shown separately; no causal claim.	Asked daily, weekly or monthly, then full Monday–Sunday weeks. Every figure exact; unknown kept separate; warned the weeks aren't adjacent; "arithmetic contributions, not evidence of cause".	Pass	not rerun
Q04 · Comparison of two metrics *Did engaged users grow with active users?* Runtime: Codex Q04/codex-run-001	MEU 4,873 → 6,734 (+38.2%) vs MAU +44.2%; engagement rate 61.4% → 58.8%; 60-second rule with watch time allocated to the month.	All six months exact; threshold and allocation stated; paid-social engagement split shown as an association.	Pass	not rerun
Q05 · Filter; overlapping slices *Did the platform mix change?* Runtime: Codex Q05/codex-run-001	No material change. Sep: Android 6,312, iOS 5,237, web 3,140 of 11,447 MAU; Aug 6,022 / 4,926 / 3,041; about 28% of users on 2+ platforms, so platform counts can't be added.	Aug vs Sep exact, shares and overlap (28.5%, 28.2%) exact; explained that rows can't be summed; stated its comparison months.	Pass	not rerun
Q06 · Lifecycle (growth accounting) *Are more users new, retained or returning?* Runtime: Codex Q06/codex-run-001	May–Sep new / retained / returning, e.g. May 2,201 / 4,757 / 2,176 and Sep 1,720 / 5,071 / 4,656; April can't be classified; groups reconcile to MAU; no causal language.	Every month exact with shares of MAU; April excluded with the reason; reconciled. Headline said retained is largest without stressing that returning users doubled.	Pass	not rerun
Q07 · Cohort conversion *Did newer signup cohorts convert to subscription at a lower rate?* Runtime: Codex Q07/codex-run-001 Q07/codex-run-002	Yes: May–Jun 1.94% (94 / 4,854) vs Jul–Aug 0.53% (29 / 5,426). April flagged as the start-up month (no subscribers at the start); September excluded (window incomplete).	Round 1: rates exact but April pooled into the baseline (Apr–Jun 2.59%), overstating the drop (79%). Round 2: May–Jun baseline, exact, April shown separately.	Fail	Pass
Q08 · Simple aggregate (revenue) *How much net revenue came from subscriptions versus PPV?* Runtime: Codex Q08/codex-run-001	Apr–Sep: subscription $59,900.04, PPV $411.93, total $60,311.97 (PPV 0.68%); gross vs net; refunds to the original type; unattributed refunds reported even at $0.	Exact to the cent with gross, refunds and net by type; booked at payment success; "zero unattributed refunds".	Pass	not rerun
Q09 · Time comparison *Did refunds materially change the revenue comparison?* Runtime: Codex Q09/codex-run-001 Q09/codex-run-002	No. Gross $30,182.93 → $30,192.92; refunds $37.93 → $25.95; net $30,145.00 → $30,166.97. Refunds change the size of a tiny increase, not the conclusion.	Exact; same conclusion; sensitivity on the 13 corrected payments. Round 2 also flagged April and added a fair May–Jun vs Aug–Sep comparison (gross −1.466%, net −1.409%, both verified).	Pass	Pass
Q10 · Ambiguous definition *How many subscription cancellations took effect during a period?* Runtime: Codex Q10/codex-run-001	State the meaning: access actually ended vs a cancellation request. August: 113 access losses, 92 requests. Use subscription history, not the user snapshot.	Gave both (113 ended, 92 requests) with exact August boundaries from history; Bronze recount agreed.	Pass	not rerun
Q11 · Funnel *Where do users drop from viewing a paywall to starting checkout and purchasing?* Runtime: Codex Q11/codex-run-001 Q11/codex-run-002	Biggest drop before checkout. Subscription May–Sep: 39.4% start checkout, 27.6% of checkouts purchase; rates flat from May; April is the start-up month; link gaps reported.	Main answer, unit and link gaps right (175 checkouts without a paywall link; 56 purchases without a checkout link). In both rounds read April's start-up as a funnel decline and pointed at a "May transition".	Partial	Partial
Q12 · Ambiguous measure; data trap *Which episodes are watched most, and how much is actually completed?* Runtime: Codex Q12/codex-run-001	State the measure. Top: title c01 ep 3, 3,738 viewers and 8,014 views; completion 28.3%; overall 33.0% (35,017 / 105,989). Trap: premium c03 ep 13, 0 of 1,455 completed.	Ranked by viewers and said so; every figure exact; flagged the 0-of-1,455 episode as unexplained without claiming a cause.	Pass	not rerun

Product Analytics Agent

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Test	Expected (answer key)	Agent result	R1	R2
Q13 · Data-quality trap *Is an unusually large source payment real?* Runtime: Codex Q13/codex-run-001	No. Raw 999.0 USD on 7 payments is a reviewed 100× unit error; correct amount $9.99. Using raw values would overstate revenue by $6,923.07. Review time unknown; correction came from the review, not new detection.	Exactly as expected, including the $6,923.07 impact and unknown review time; raw, curated and review shown separately.	Pass	not rerun
Q14 · Missing data *Can we trust acquisition-channel conclusions when source values are missing?* Runtime: Codex Q14/codex-run-001	Yes at channel level: unknown is 135 of 13,600 signups (1.0%) and 238 users; kept separate; a sensitivity bound; no causal or campaign attribution.	Exact; showed the ranking holds even with all unknowns assigned to one channel; no causal claim.	Pass	not rerun
Q15 · Vague question (should clarify) *"Performance is down. What happened?"* Runtime: Codex Q15/codex-run-001	Ask which metric and period before analysing. For the agreed scope (net revenue, Sep vs Aug): $9,819.93 → $10,175.51 (+3.62%).	Asked first; for the agreed scope every figure exact; showed the premise wasn't supported and suggested confirming the KPI.	Pass	not rerun
Q16 · Unanswerable (should decline) *Which advertising campaign caused each user's purchase?* Runtime: Codex Q16/codex-run-001	Decline: there is no campaign-level exposure or user link; explain what can be compared and what data is needed; no fabricated join.	Declined; showed there are no campaign fields; listed the data needed (campaign IDs, click events, holdouts).	Pass	not rerun
Q17 · Filter (subscribers by country) *How many US users had an active subscription at the end of September?* Runtime: Codex Q17/codex-run-001	349 at 1 Oct 2026 00:00 UTC: 312 renewing plus 37 cancelled but paid up; the user's country.	349 exact, with the 37 explained; history recount also 349.	Pass	not rerun
Q18 · Filter; time comparison *How many mobile sessions were there in September versus August?* Runtime: Codex Q18/codex-run-001	Define mobile (iOS + Android). 36,295 → 38,366 (+5.7%) counting sessions that touch each month; same counting in both months.	Asked what mobile means; August exact; September 38,365 by start month and 38,366 by overlap, both stated.	Pass	not rerun

04 What failed, why, and what changed

Failure	Where	What changed	Result
Conclusions firmer than the evidence: a cause stated from an association; two hypotheses rejected too firmly; data-quality impact called "none"	Q02 v1	Skill rules: separate pattern from cause; quantify data-quality scope and impact or say "unknown"; keep unexplained anomalies open	Not seen in the 17 Codex runs; the blind Q02 rerun passed but still worded two data-quality impacts a little too firmly
Comparison periods mislabelled; a monthly amount compared with quarterly revenue	Q02 v1	Skill rule: exact periods, the same time basis, explicit start and end boundaries	Not seen again
The first month of data used as a normal baseline (the subscriber base starts at zero in April)	Q07, Q11	Skill rule: check whether the first period is a start-up period before using it as a baseline	Q07 fixed and Q09 applied it correctly; Q11 unchanged because the question isn't phrased as a comparison
A harmless registry-version warning repeated in every report	All runs	Verified the change since the build is documentation only (all 56 SQL definitions identical) and recorded it	Still raised in 3 of 4 Codex round-2 runs; the blind Q02 run read the note and handled it correctly

Hardest question type: comparisons that need context the data doesn't announce, such as a start-up month. The agent computes correctly but only questions the fairness of a comparison when a rule names the situation.

05 The GM question in detail (Q02)

v1 (round 1)	The agent's own report in Claude Code. Numbers and story right; scored Fail under the strict rule for overstated conclusions, mislabelled periods and a monthly/quarterly mix-up.
Review	A reviewer checked v1 against the evidence and source data: seven substance points plus format feedback, kept word for word.
v2	The agent's revision after that review: every point addressed and confirmed by five extra checks (17–21). This is the board report in the submission.
Blind rerun	A fresh Claude Code agent on the updated skill, given only the standard prompt and a clean repository copy (no review, answer key or earlier reports; told not to read anything else): numbers exact, material review points fixed, start-up month handled; two data-quality statements and one rejected hypothesis still a little too firm. Pass with minor issues.

06 Where to find everything in the repository

Repository: https://github.com/SoumabrataSarkar/product-analytics-agent (private; reviewers need collaborator access). To get a local copy: git clone https://github.com/SoumabrataSarkar/product-analytics-agent.git

Path	What it holds
submission/	Start here: board report (v2), the agent's raw v1, one-page write-up and this file
docs/evaluations/results.md	The same results as a page in the repository
docs/evaluations/investigation-paths.md	The 18 scenarios and what success requires
evaluation-results/answer-key/	Expected values and scoring rules (batches 1–4) and the reference query logs
evaluation-results/QNN/codex-run-001/	Round 1 run for question NN: scoring record, the agent's output, evidence zip
evaluation-results/QNN/codex-run-002/	Round 2 reruns (Q01, Q07, Q09, Q11)
evaluation-results/Q02/run-001/	GM question: v1, review feedback word for word, v2, list of v1 → v2 changes
evaluation-results/Q02/run-002-blind/	Blind rerun of the GM question on the updated skill
evaluation-results/TRACKER.md	Status of every question in both rounds
skills/product-analytics-investigation/SKILL.md	The shared skill, including the rules added after each failure

Agents must not read evaluation-results/, submission/ or docs/evaluations/ during an investigation (AGENTS.md), because they contain expected answers.